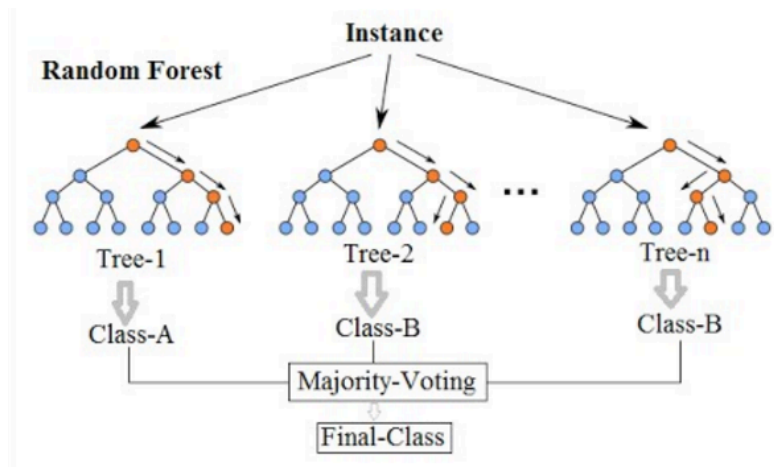


Random Forest Output Explanation - M.L. in Finance

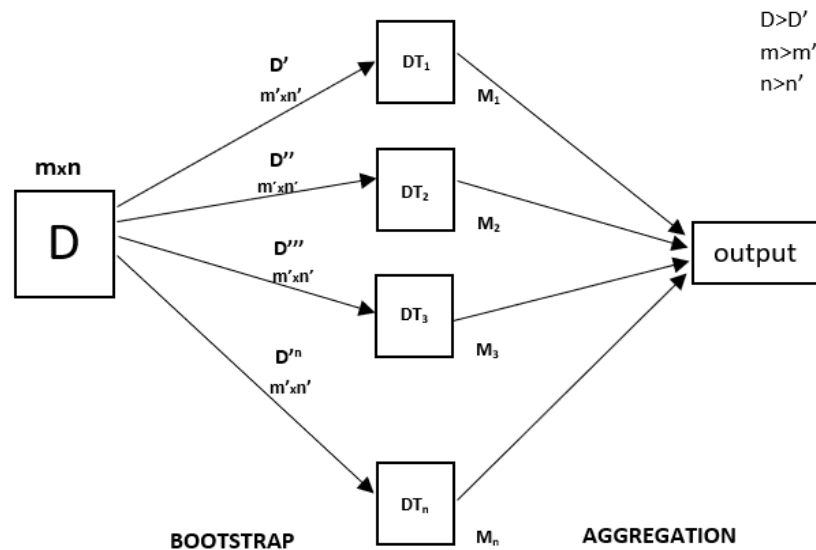
Machine Learning Methodology :

The machine learning technique I am using is called a Random Forest. It is a method that combines the predictions from multiple decision trees to produce a more accurate and stable prediction. Whether it is for classification or regression.

For predicting data, we need to use a Random Forest Regression. It works by creating multiple decision trees, each trained on a random subset of the data. The process begins with bootstrap sampling, where random rows of data are selected with replacement to form different training datasets for each tree. After this, we perform feature sampling, where only a random subset of features is used to build each tree, ensuring diversity in the models.



After the trees are trained, each tree makes a prediction, and the final prediction for the regression task is the average of all the individual tree predictions; this process is called aggregation.



In short, a random forest is very useful in order to predict with accuracy a target data with large and different types of data, as it captures well the relationships between them. It is also less prone to overfitting thanks to how it handles the learning process; it generalizes a rule for the data rather than 'memorizing it' like most machine learning models.

Key Hyperparameters :

- n_estimators: Number of decision trees in the model. More trees generally improve performance but increase computation time.
- max_depth: Ensures that each tree has a maximum depth. This hyperparameter is important since it controls complexity; deep trees will tend to overfit.
- min_samples_leaf: Minimum number of samples that must be present in a leaf node. Ensures that the tree doesn't create nodes with very few samples, which could lead to overfitting.
- max_features: Number of features to consider when splitting a node. Determines the maximum number of features to be considered for each tree.
- min_samples_split: Minimum number of samples required to split an internal node. Ensures that nodes with fewer samples are not split, helping to keep the tree simpler and more general. It reduces the risk of overfitting.
- bootstrap: Whether to build trees on random subsets (with replacement) of the training data (True) or the entire dataset (False)
- max_samples: Maximum number of samples to draw from the dataset to train each base estimator when bootstrap=True. Limiting the number of samples per tree

speeds up the training process but may reduce accuracy, as each tree is trained on a subset of data.

- max leaf nodes: Limits the number of leaf nodes in the tree, hence controlling its size and complexity.

Interpretation of the Random Forest results :

1) Random Forest Prediction Metrics :

Out-of-Bag Score = 0.989

The Out-of-Bag score is an internal validation metric, an unbiased estimate of model performance using data not seen by individual trees during their bootstrap training, acting like an automatic cross-validation. A high OOB score means that our model generalizes well, showing strong performance (high accuracy/low error).

MSE = 0.296

R2 = 0.769

Adjusted R2 = 0.741

The calculation of the Mean Squared Error is quite low; our predictions are close predictions to the actual data. R2 and Adjusted R2 scores tell the proportion of dependent variable variance explained by all predictors. R2 always increases with more predictors, but Adjusted R2 only increases if the new predictor genuinely improves the model. A value of 76.9% means that 76.9% of the variation in the data is explained by the model, while the remaining 23.1% is due to other factors. This means there is still some room for improvement, but it is quite good for a model in practice.

2) Feature Importance Table :

US GDP	25.77%
US T-Note 5Y Yield	15.89%
Interest Rates	14.46%
US T-Note 2Y Yield	12.91%
XAU/USD	9.20%
US T-Bond 30Y Yield	8.57%
OIL WTI	6.65%
US Unemployment	3.60%
USD/EUR	1.45%

USD Index	1.08%
VIX	0.41%

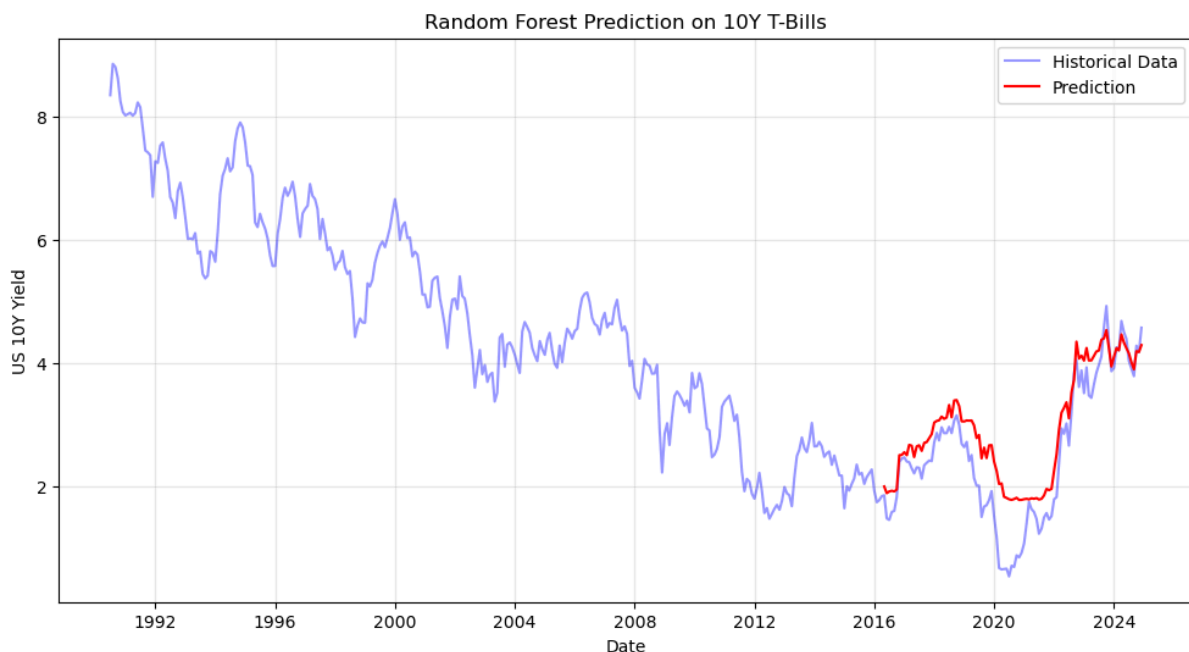
This feature importance table summary shows us which type of data contributed the most to the least to the model prediction.

Without surprise, the US T-note YTM for 5-year and 2-year maturities are among the most influential features (with 28.8% importance combined) for the prediction of the 10-year US T-note YTM. A similar expectation with the Interest Rates being the 3rd most important feature. Interest rates influence the number of bank loans allocated and the internal consumption rate. It is known that interest rates have an inverse correlation with bond prices.

On the contrary, it is surprising that the US GDP counts for more than a quarter of the importance for the model. Strong GDP growth leads to higher interest rates, which leads to a drop in current bond prices. Overall, the model captured well the indirect relationship between the US 10Y T-Note YTM and economic health and policies.

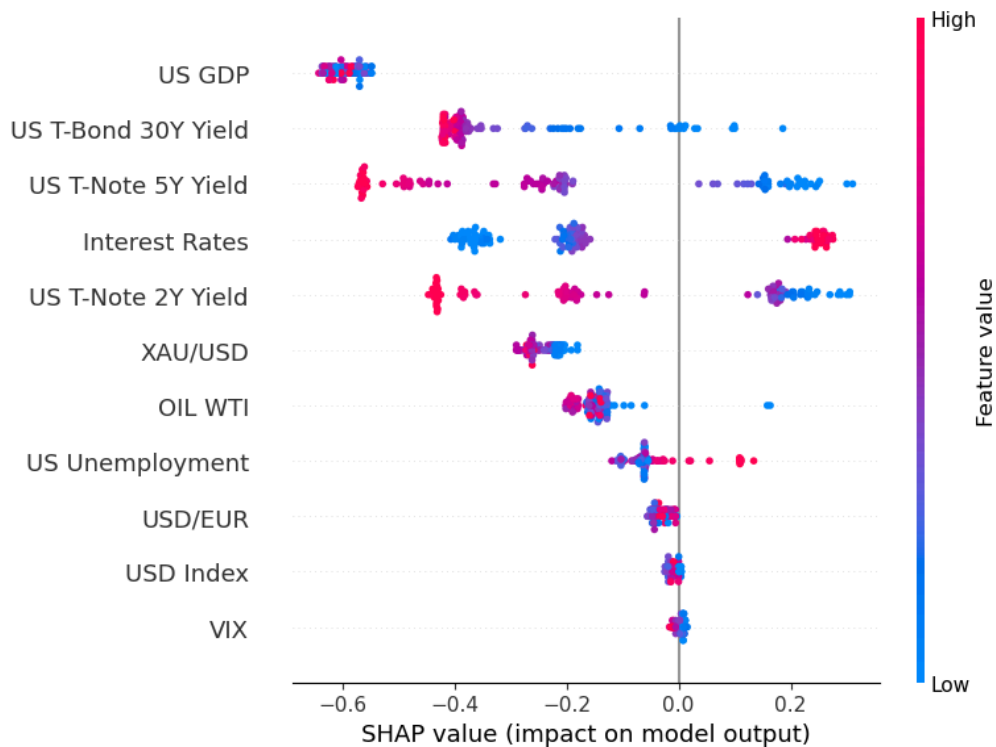
The rest of the features are not that relevant for the model, except for the spot price of gold, which ranks 5th with an importance of 9.20%. The 30Y T-bond YTM doesn't seem to affect our target a lot.

3) Actual Data VS Prediction plot :



Here we can see the plot of our prediction over 9 years alongside the full historical data. For the first 3 to 4 years, the prediction seems to be doing fine; it is a little bit overestimating the yield. For the COVID-19 period, the prediction has anticipated a drop. However, it couldn't predict the much larger steepness from the impact of COVID-19 and the change of the regime that it had. After this, it starts to predict the historical data with accuracy.

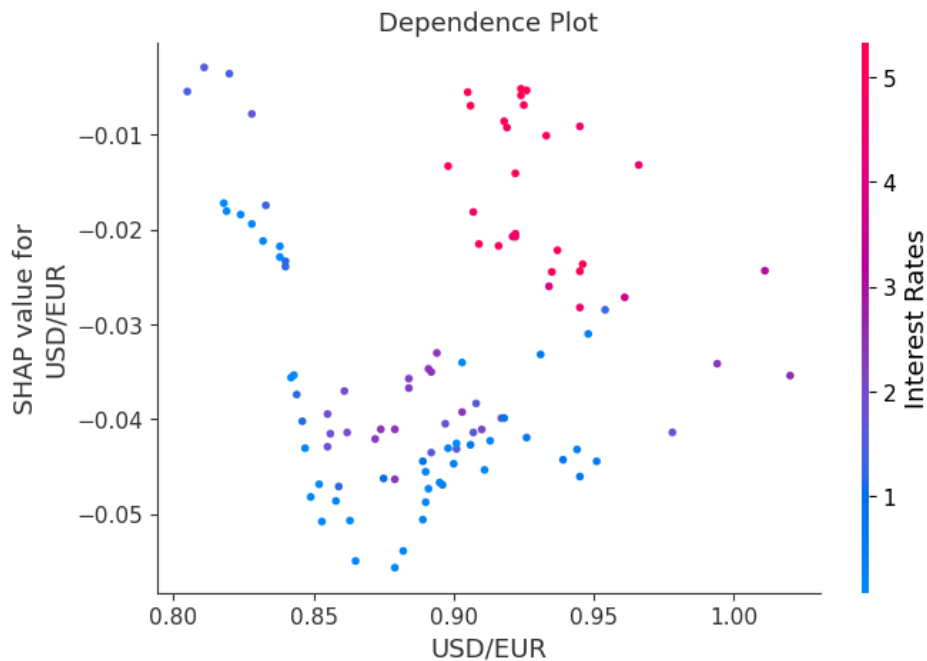
4) SHAP :



This graph, outputted by performing SHapley Additive exPlanations, shows us how our features impact the model. Red bars push the prediction higher, blue bars push it lower, and the bar length shows the feature's impact.

As we already saw in the feature importance table, the bottom tier bars are closer to the middle with a roughly equivalent mix of red and blue. This confirms their low importance.

We again find the interest rates that have a long bar, but most importantly we can see that $\frac{2}{3}$ of the bar is blue since the interest rate was quite high in the past, and $\frac{1}{3}$ of it is red since it was quite low recently.



This is a SHAP dependence plot; it shows interaction effects with other features. Meaning that an instance's SHAP value for a feature is not solely dependent on the value of that feature, but also influenced by values of the instance's other features.

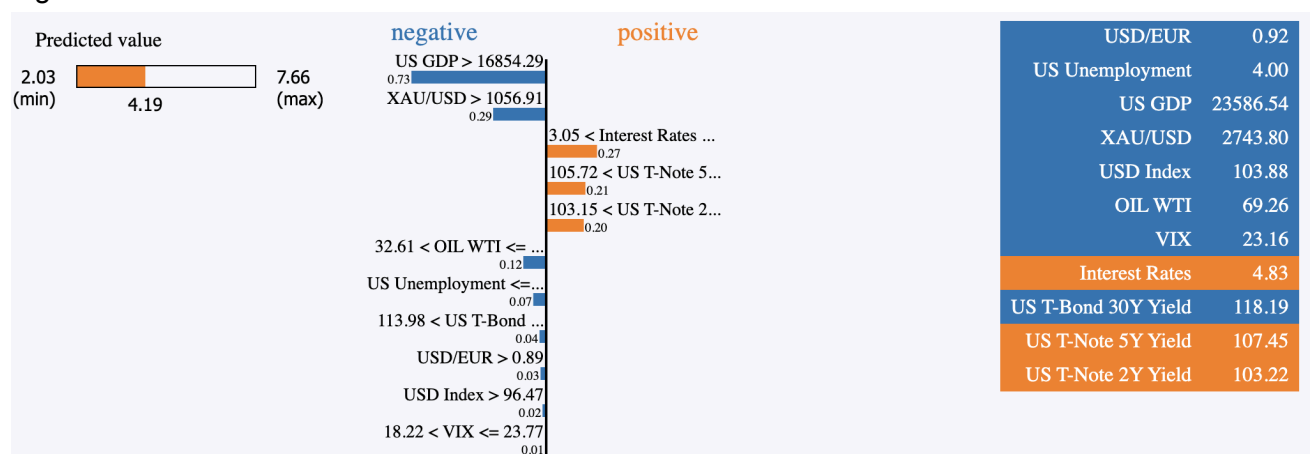
This plot shows that the foreign exchange USD/EUR happens to have the largest interaction effect with interest rates.

5) LIME :

Intercept : 4.85

Prediction_local : [4.24]

Right : 4.19



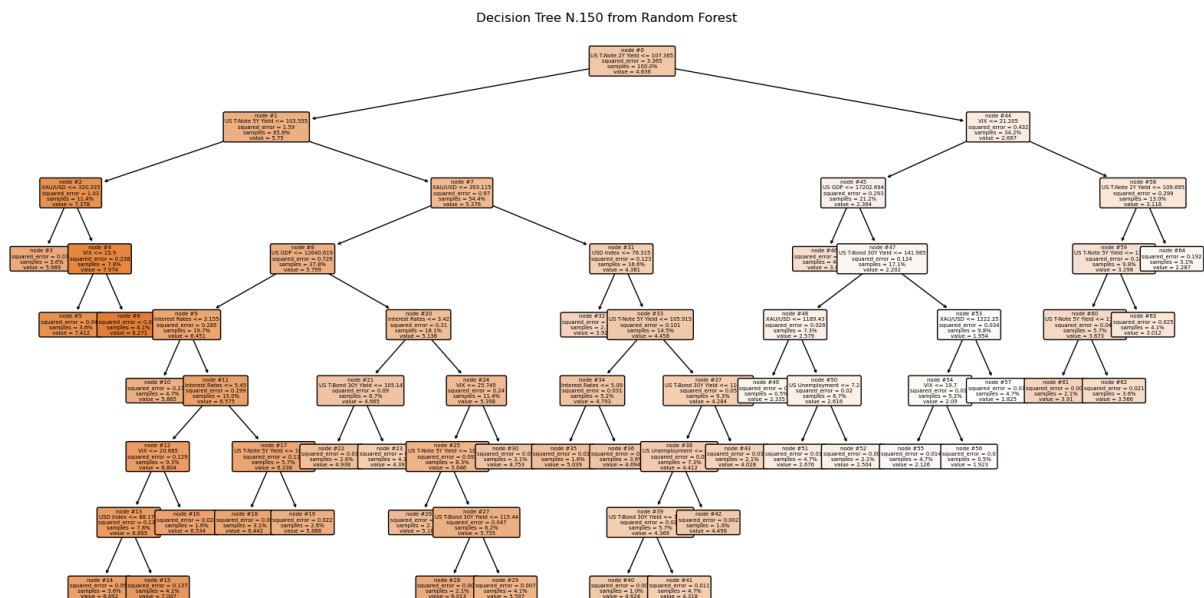
LIME is a process for understanding which features push a model's prediction towards a certain outcome for a specific data point, visualized as bars showing feature importance

(positive/negative impact) on a local model, helping you see why the 'black-box' model made that decision.

Interest Rates, US T-Note 5Y and 2Y have the largest positive weight, suggesting those were the most significant factors pushing the prediction positively.

US GDP and Gold spot price have the largest negative weight, indicating those feature values pushed the prediction negatively.

6) Tree N.150 of the Random Forest :



Here we can see one of the decision trees from the random forest; it doesn't tell us much about the prediction or features' impact. However, it is useful when tuning hyperparameters; we can see the depth, leaves, and nodes of the trees. Bad hyperparameter tuning will show a way more complicated tree with too few or too high number of leaves and/or node splits.